

Assignment2 – Bayesian Networks

Given: April 29 Due: May 4

Problem 2.1 (Probabilities in Python)

Complete the partial *implementation* of probabilities at <https://kwarc.info/teaching/AI/resources/AI2/probabilities/>

Problem 2.2 (Bayesian Rules)

Give the formulas and a one-sentence explanation of the following basic rules in Bayesian inference:

1. Bayes rule
2. Product rule
3. Chain rule
4. Marginalization
5. Normalization

Problem 2.3 (Is your TA in the office?)

You want to discuss something with your TA. You know that

1. the probability of your TA being in the office, assuming it is morning, is $\frac{1}{5}$,
2. if your TA is in the office, there is a $\frac{1}{3}$ probability it is morning,
3. the probabilities that it is morning or afternoon are both $\frac{1}{2}$

Your tasks:

1. Write down the probabilities mentioned above as formulas
2. Compute the full joint probability distribution
3. What's the probability you'll meet your TA, if you come to the office in the afternoon?

Problem 2.4 (AFT Tests)

Trisomy 21 (*Down syndrome*) is a genetic anomaly that can be diagnosed during pregnancy using an amniotic fluid test.

The probability of a foetus having Down syndrome is strongly correlated with the age of the pregnant parent. We will only consider the following two age groups.

1. For 25 year olds the probability is one in 1250,
2. for 43 year old parents it increases to one in fifty.

However, diagnostic tests are never perfect. We distinguish two kinds of errors:

3. Type I Error (False Positive): The test result is positive even though the child is healthy.
4. Type II Error (False Negative): The test result is negative even though the child has trisomy 21.

The probabilities of Type I and Type II Errors are both merely 1% for amniotic fluid tests for Down syndrome.

1. Express the four items above in the form of conditional probabilities. Use the random variable F with domain $\{Age_{25}, Age_{43}\}$ for the age of the pregnant person and the Boolean random variables Pos and $Down$ for the propositions “The amniotic fluid test is positive” and “The child has Down syndrome” respectively.
2. Assume that we have a 25 year old pregnant person. Using Bayes’ theorem, express and compute the probability that their child has Down syndrome, given that the amniotic fluid test is positive. What can we conclude from the result?

Problem 2.5 (Bayesian Networks)

Consider the following situation about an alarm clock:

- It fails to ring if its batteries are empty.
- It fails to ring if it the volume is muted.
- You might oversleep if your alarm clock fails.
- You might oversleep if you stay up late.

You want to *model* this situation as a *Bayesian network* using *Boolean random variables*.

1. Give an appropriate *set of random variables* and their *meaning* and draw the *Bayesian network* (using an appropriate *variable order*).
2. Give the *probability* of the clock failing in terms of the entries of the *conditional probability tables* of your *network*.

Now you decide to *model* the failing of the clock as a *deterministic node*.

3. Explain (in about 3 *sentences*) whether that decision is justified by the description, and how it changes the *conditional probability tables* of your *model*.
4. State the *probability* of the clock failing in terms of the entries of the *conditional probability tables* of your *network* (with the *deterministic node* for the failing clock).